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School of Engineering and Technology, Civil Engineering Department
Formative Assessment – I (4th Semester)
CE-BCE23404T – Hydraulic Engineering

27th March, 2026
Duration: 1 hour

Total Marks: 20

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1. Fill in the blanks: (5)
- (i) The expression for shear stress in turbulent flow is
 - (ii) When $k > \delta$, boundary is called.....
 - (iii) The expression for Eddy viscosity is
 - (iv) When the Reynold's number is more than 5×10^5 , the boundary layer is called
 - (v) The components of the total force in the direction of motion is called
2. A pipe-line carrying water has average height of irregularities projecting from the surface of the boundary of the pipe as 0.20 mm. what type of the boundary is it? The shear stress development is 7.848 N/m^2 . Take value of kinematic viscosity for water as 0.01 stokes. (5)
3. Define Displacement Thickness and derive the expression for it. (10)